

1. Given the following list:

None

```
List<Integer> numbers = Arrays.asList(10, 25, 30, 45, 50, 65, 70, 85, 90);
```

- Find numbers greater than 40.
- Keep only even numbers.
- Square them.
- Sort them in descending order.
- Display the result.

2. Given:

None

```
List<Integer> numbers = Arrays.asList(10, 50, 30, 80, 50, 90, 70, 90, 60);
```

- Remove duplicate values.
- Sort in descending order.
- Find the second-highest number.
- Display the result.

3. Given:

None

```
List<String> names = Arrays.asList(  
    "Ravi", "Anil", "Suresh", "Arun",  
    "Ravi", "Mahesh", "Ajay", "Suresh"  
);
```

- Remove duplicate names.
- Find names starting with "A".
- Convert them to uppercase.
- Sort alphabetically.
- Display the result.

4. Given:

None

```
List<Integer> numbers = Arrays.asList(10, 20, 30, 40, 50, 60, 70);
```

- Calculate the average.
- Find numbers greater than the average.
- Sort them in descending order.
- Display the result.

5. Given:

None

```
List<Integer> numbers = Arrays.asList(12, 24, 36, 48, 60);
```

- Check whether any number is greater than 50.
  - Check whether all numbers are divisible by 12.
  - Check whether none of the numbers is negative.
  - Display all three results.
6. Create an Employee class with id, name, department, and salary. Given a list of employees:

None

|     |        |       |       |
|-----|--------|-------|-------|
| 101 | Ravi   | IT    | 60000 |
| 102 | Anil   | HR    | 45000 |
| 103 | Suresh | IT    | 80000 |
| 104 | Priya  | Sales | 55000 |
| 105 | Arun   | IT    | 70000 |

- Find employees whose salary is greater than ₹50,000.
  - Display only their names.
  - Sort the names alphabetically.
7. Given a list of Employee objects:

None

|        |       |
|--------|-------|
| Ravi   | 60000 |
| Anil   | 45000 |
| Suresh | 90000 |
| Priya  | 75000 |
| Arun   | 90000 |

- Find the highest salary.
  - Find the second-highest unique salary.
  - Display the employee(s) having the second-highest salary.
8. Given a list of Employee objects belonging to:

None

IT  
HR  
IT  
Sales  
HR  
IT  
Sales  
Testing

- Display each department.
  - Display the number of employees in each department.
  - Find the department having the highest number of employees.
9. Given:

None

|        |       |       |
|--------|-------|-------|
| Ravi   | IT    | 60000 |
| Anil   | HR    | 45000 |
| Suresh | IT    | 80000 |
| Priya  | HR    | 55000 |
| Arun   | Sales | 70000 |
| Kiran  | IT    | 90000 |

- Calculate the average salary of each department.
- Display department and average salary.
- Find the department having the highest average salary.

10. Given:

None

|        |         |       |
|--------|---------|-------|
| Ravi   | IT      | 60000 |
| Anil   | HR      | 45000 |
| Suresh | IT      | 90000 |
| Priya  | Sales   | 55000 |
| Arun   | IT      | 70000 |
| Kiran  | Testing | 65000 |

- Select employees from IT or Testing.
- Salary must be greater than ₹60,000.
- Sort them by salary in descending order.
- Skip the highest-paid employee.
- Display the next 2 employees.
- Check whether any employee has a salary greater than ₹80,000.